

Step 1 – Update on b_i and $W_i^{(1)}$

Gradients depend on loss through $\hat{f}(x)$. Let $\delta = \hat{f}(x) - y$.

From backprop:

$$\frac{\partial \ell}{\partial b_i} = \delta \cdot W_i^{(2)} \cdot \mathbf{1}_{z_i > 0}, \quad \frac{\partial \ell}{\partial W_i^{(1)}} = \delta \cdot W_i^{(2)} \cdot x \cdot \mathbf{1}_{z_i > 0}.$$

Step 2 – Updated parameters

$$b'_i = b_i - \lambda \frac{\partial \ell}{\partial b_i}, \quad W_i^{(1)'} = W_i^{(1)} - \lambda \frac{\partial \ell}{\partial W_i^{(1)}}.$$

Step 3 – New elbow

$$e'_i = -\frac{b'_i}{W_i^{(1)'}}.$$

Substitute updates to express in terms of old parameters, δ , λ , x , and indicator.

Final expression:

$$e'_i = -\frac{b_i - \lambda \delta W_i^{(2)} \mathbf{1}_{z_i > 0}}{W_i^{(1)} - \lambda \delta W_i^{(2)} x \mathbf{1}_{z_i > 0}}.$$

Uncertainty note:

In 5(b)(ii)/(iv), numerical results depend on specific numbers; I gave examples to illustrate.

In 5(d), the full derivation assumes ReLU derivative at 0 is 0 (as given).

All reasoning steps are shown as requested.

Map out every concept I should know or review before working on this assignment. For example, create a prerequisite knowledge map for fully understanding Ridge Regression as MAP estimation, including linear algebra, probability, and optimization. For each learning module, list the key concepts, common misunderstandings, and 3 exercises to build those skills.

Prerequisite Knowledge Map for EECS 182 Homework 0

I. VECTOR CALCULUS & LINEAR ALGEBRA FOUNDATIONS

Key Concepts:

1. Vector/Matrix Derivatives

- Scalar-by-vector derivative $\rightarrow$ row vector (Jacobian convention)
- Vector-by-vector derivative $\rightarrow$ matrix
- Chain rule for multivariable functions
- Quadratic forms: $\partial(x^T A x)/\partial x = x^T(A + A^T)$

2. SVD and Matrix Properties

- Full vs. reduced SVD
- Orthogonal matrices: $U^T U = I$, $V^T V = I$
- Σ as rectangular diagonal matrix
- Rank and singular values

3. Matrix Inverses and Pseudo-inverses

- Left inverse: $A^+ A = I$ (tall matrices)
- Right inverse: $A A^+ = I$ (wide matrices)
- Moore-Penrose conditions

Common Misunderstandings:

- Confusing numerator-layout vs denominator-layout conventions
- Misapplying chain rule with matrix dimensions
- Forgetting that Σ^+ uses reciprocal singular values only for nonzero σ_i
- Assuming $(AB)^{-1} = B^{-1}A^{-1}$ without checking invertibility

Skill-Building Exercises:

1. Derive $\partial(x^T M x)/\partial x$ for symmetric and non-symmetric M
2. Compute SVD of $[[1,2],[2,1],[0,0]]$ and write explicit Σ^+
3. Show that $(A^T A)^{-1} A^T$ is a left inverse when A has full column rank

II. LEAST SQUARES & OPTIMIZATION THEORY

Key Concepts:

1. Overdetermined Systems ($m > n$)

- Normal equations: $X^T X w = X^T y$
- Geometric interpretation: projection onto column space
- Connection to SVD: $w^* = V \Sigma^+ U^T y$

2. Underdetermined Systems ($m < n$)

- Min-norm solution: $w^* = X^T (X X^T)^{-1} y$
- Null space of X
- Regularization motivation

3. Ridge Regression Formulations

- Optimization: $\min \|y - Xw\|^2 + \lambda \|w\|^2$
- SVD interpretation: $\sigma_i / (\sigma_i^2 + \lambda)$ factors
- Limits: $\lambda \rightarrow 0$ (OLS/min-norm), $\lambda \rightarrow \infty$ (zero weights)

Common Misunderstandings:

- Assuming OLS works for wide matrices (it doesn't without modification)
- Confusing min-norm with regularized solutions
- Misinterpreting λ as "small" vs "large" relative to σ_i^2
- Forgetting that ridge always has unique solution ($X^T X + \lambda I$ invertible)

Skill-Building Exercises:

1. Given $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $y = [1, 3, 2]$, find OLS and ridge ($\lambda=1$) solutions
2. Show that min-norm solution has no component in $\text{null}(X)$
3. Derive ridge solution from completing the square in w

III. PROBABILITY & STATISTICAL INTERPRETATIONS

Key Concepts:

1. Multivariate Gaussian Distribution

- PDF: $p(x) \propto \exp(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu))$
- Properties: linear transformations remain Gaussian
- Independence vs. uncorrelated for Gaussians

2. Bayesian Inference

- Prior: $p(w)$
- Likelihood: $p(y|w, X)$
- Posterior: $p(w|y, X) \propto p(y|w, X)p(w)$
- MAP estimation: $\text{argmax}_w p(w|y, X)$

3. Ridge as MAP with Gaussian Priors

- Prior: $w \sim N(0, I/\tau)$ or $N(0, I)$
- Likelihood: $y|w \sim N(Xw, \sigma^2 I)$
- Posterior mean = ridge solution with $\lambda = \sigma^2/\tau$

Common Misunderstandings:

- Confusing MAP with MLE (MLE is $\lambda=0$ case)
- Misplacing variance parameters in Gaussian PDF
- Forgetting that ridge assumes zero-mean prior
- Not recognizing λ as noise-to-prior variance ratio

Skill-Building Exercises:

1. Show $N(\mu, \sigma^2)$ PDF normalized correctly (integral = 1)
2. Given prior $w \sim N(0, 1)$, likelihood $y|w \sim N(w, 4)$, find MAP for observed $y=3$
3. Derive ridge from $y = Xw + \varepsilon$, $w \sim N(0, \gamma I)$, $\varepsilon \sim N(0, \sigma^2 I)$

IV. NEURAL NETWORK MECHANICS

Key Concepts:

1. ReLU Activation & Piecewise Linearity

- $\phi(z) = \max(0, z)$
- Gradient: 1 if $z > 0$, 0 if $z < 0$, convention at 0
- "Elbow" at $z=0$

2. Backpropagation Through Single Neuron

- Chain rule: $\partial \ell / \partial w = (\partial \ell / \partial \phi) \times (\partial \phi / \partial z) \times (\partial z / \partial w)$
- Indicator function from ReLU derivative
- Weight and bias updates

3. Gradient Descent Dynamics

- Update rule: $\theta' = \theta - \eta \nabla \ell(\theta)$
- Learning rate η effects
- Movement of decision boundaries (elbows)

Common Misunderstandings:

- Taking gradient through ReLU at 0 (ambiguous, convention needed)
- Confusing pre-activation z with activation $\phi(z)$
- Forgetting indicator in ReLU gradient
- Misunderstanding how elbow location changes with updates

Skill-Building Exercises:

1. Compute gradients for $\ell = \frac{1}{2}(\text{ReLU}(wx+b)-y)^2$ for cases: $wx+b>0$, <0 , $=0$
2. For $w=2, b=-1, x=1, y=3, \eta=0.1$, compute one GD update and new elbow
3. Show that if $\phi(x)=0$ and indicator=0, elbow doesn't move even if prediction wrong

V. MATRIX IDENTITIES & PROOF TECHNIQUES

Key Concepts:

1. Woodbury Identity

- $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$
- Special case: $(X^T X + \lambda I)^{-1} X^T = X^T (X X^T + \lambda I)^{-1}$

2. Proof Strategies

- Component-wise derivation vs. matrix manipulation
- Verifying dimensions at each step
- Checking special cases

3. Limit Analysis

- $\lambda \rightarrow 0$: connecting ridge to OLS/min-norm
- $\lambda \rightarrow \infty$: shrinkage to zero
- Relative scale: $\sigma_i \ll \lambda$ vs $\sigma_i \gg \lambda$

Common Misunderstandings:

- Applying Woodbury when dimensions don't match
- Not checking invertibility conditions
- Taking limits without considering matrix rank
- Confusing asymptotic behaviors

Skill-Building Exercises:

1. Prove $(X^T X + \lambda I)^{-1} X^T = X^T (X X^T + \lambda I)^{-1}$ using Woodbury
2. Analyze $\lim_{\lambda \rightarrow 0} (X^T X + \lambda I)^{-1} X^T$ for tall vs. wide X
3. Show $\Sigma^+ = \lim_{\lambda \rightarrow 0} (\Sigma^T \Sigma + \lambda I)^{-1} \Sigma^T$

VI. IMPLEMENTATION & NUMERICAL CONSIDERATIONS

Key Concepts:

1. Numerical Stability

- Small singular values $\rightarrow$ large errors in pseudo-inverse
- Ridge as numerical stabilization
- λ as minimum singular value threshold

2. Computational Approaches

- Solving normal equations vs. SVD
- When to use each method
- Cost considerations

3. Practical Regularization

- Choosing λ via cross-validation
- Interpreting λ in context
- Standardization effects

Common Misunderstandings:

- Thinking OLS always computable (ill-conditioned $X^T X$)
- Believing SVD always better (computationally expensive)
- Using same λ for unstandardized features
- Ignoring numerical precision issues

Skill-Building Exercises:

1. For ill-conditioned X , compare OLS and ridge solutions numerically
2. Implement both normal equations and SVD ridge solutions
3. Show that standardizing features changes optimal λ

LEARNING PATH RECOMMENDATION:

Week 1: Vector calculus + Linear algebra review

Exercises: Matrix derivatives, SVD practice, dimension checking

Week 2: Least squares theory + Optimization

Exercises: Normal equations derivation, SVD solutions, min-norm proofs

Week 3: Probability + Bayesian interpretation

Exercises: Gaussian properties, MAP derivation, λ interpretation

Week 4: Neural network mechanics

Exercises: ReLU gradients, backprop through layers, elbow dynamics

Week 5: Advanced proofs + Applications

Exercises: Woodbury identity, limit analysis, numerical implementation

CRITICAL INSIGHTS TO REMEMBER:

1. **Ridge Regression Unification:** Optimization, Bayesian, and algorithmic perspectives all converge to same solution
2. **SVD as Rosetta Stone:** Translates between algebraic, geometric, and statistical views

3. **Regularization Spectrum:** λ interpolates between OLS/min-norm ($\lambda \rightarrow 0$) and zero ($\lambda \rightarrow \infty$)
4. **ReLU Dynamics:** Gradient updates move both slope and decision boundary (elbow)
5. **Matrix Calculus Consistency:** Convention matters, but physics (gradient descent) works regardless

This map provides the conceptual scaffolding to not just solve the homework, but deeply understand why ridge regression appears in these different forms and how neural network components behave under optimization.

Explain a core concept of this homework, assuming I have NO prior knowledge. For example, explain the MAP interpretation of Ridge Regression as if I have not taken probability theory. Explicitly state every assumption and every distribution identity you rely on. At the end, list all the mathematical tools you used so I can review them.

Explain Like I'm 5: Why Ridge Regression is Like a "Skeptical Detective"

The Core Idea (Plain English)

Imagine you're trying to predict house prices. You have some data, but not much. Ridge Regression is like a detective who:

1. **Looks at the evidence** (your data)
2. **BUT is naturally skeptical** - doesn't trust extreme explanations
3. **Prefers simpler theories** unless the data strongly suggests otherwise

The "skepticism" comes from assuming weights should be small unless proven otherwise by strong evidence. This is the MAP (Maximum A Posteriori) interpretation.

Step-by-Step Breakdown

1. The Problem We're Solving

Scenario: We want to predict house prices (\$y\$) from features like size, bedrooms, location.

Our tool: A linear model:

$$\text{Price} = w_1 \times \text{Size} + w_2 \times \text{Bedrooms} + \dots + w_d \times \text{Feature}_d$$

The \$w\$ values (weights) tell us how important each feature is:

- $w_1 = 100$ means: "Each extra square foot adds \$100 to price"
- $w_1 = 100000$ means: "Each extra square foot adds \$100,000 to price" (obviously ridiculous!)

2. Ordinary Least Squares (OLS) - The "Gullible Detective"

Without skepticism, we'd use OLS:

- Finds \$w\$ that makes predictions **match training data as closely as possible**
- Problem: With little data, OLS can find crazy \$w\$ values that fit noise perfectly
- Example: If we have 3 houses, OLS might find weights that predict those 3 perfectly but fail on new houses

3. Ridge Regression - Adding "Skepticism"

Ridge says: "I want good predictions **AND** I want reasonable weights."

Mathematically:

$$\text{Minimize: } \underbrace{(\text{Prediction Error})^2}_{\text{Fit the data}} + \lambda \times \underbrace{(\text{Size of Weights})^2}_{\text{Keep weights small}}$$

The Trade-off Knob: λ

- $\lambda = 0$: No skepticism = OLS (gullible)

- $\lambda > 0$: Some skepticism
- $\lambda = \text{huge}$: Extreme skepticism $\rightarrow$ all weights = 0 ("I don't trust any features!")

4. The Bayesian Detective: MAP Interpretation

Here's where probability enters. We're going to think like a Bayesian detective:

Assumption 1: What We Believe BEFORE Seeing Data (The Prior)

We start with a **default belief**: "Weights are probably small."

Mathematically: We assume each weight w_i is drawn from a **Normal Distribution** centered at 0.

What's a Normal Distribution?

- The bell curve you've seen
- Center = 0 means "most likely value is 0"
- Spread controlled by variance: how much we allow deviation from 0

Visual: Most weights near 0, a few might be moderate, extreme values very unlikely.

Assumption 2: How Data is Generated (The Likelihood)

When we see actual house prices, we assume:

$$\text{Actual Price} = \text{Predicted Price} + \text{Random Noise}$$

The noise is also Normal Distribution:

- Center = 0 (predictions are unbiased on average)
- Spread = how much random variation in prices

Assumption 3: Combining Beliefs with Evidence (Bayes' Theorem)

The detective's reasoning:

1. **Start with prior belief:** "Weights are probably small"
2. **See the evidence:** Actual house prices
3. **Update belief:** "Given this data, what weights are most plausible?"

Mathematically:

$$\text{Updated Belief} \propto (\text{Prior Belief}) \times (\text{How well weights explain data})$$

5. The MAP Solution: "Most Plausible Explanation"

We want the weights that are **most plausible** given both:

1. Our prior belief (weights should be small)
2. The evidence (actual prices)

Key Insight: Maximizing "plausibility" = Minimizing this sum:

$$\underbrace{(\text{Data Mismatch})^2}_{\text{Penalize bad predictions}} + \lambda \times \underbrace{(\text{Weight Size})^2}_{\text{Penalize extreme weights}}$$

Surprise: This is **exactly** the Ridge Regression formula!

The λ turns out to be:

$$\lambda = \frac{\text{Noise Variance}}{\text{Prior Variance}}$$

Interpretation:

- Lots of noise $\rightarrow$ trust data less (bigger λ)
- Strong prior belief $\rightarrow$ trust data less (bigger λ)

Concrete Example

Prior belief: "Square footage coefficient is probably around \$100/ft², almost certainly between \$50 and \$150"

We see data: One mansion suggests \$1000/ft²

Ridge/Detective thinks: "That one data point is probably noise or special case. I'll adjust toward \$100 but not all the way."

OLS/Gullible detective: "Data says \$1000/ft², so that must be right!"

Every Assumption Made Explicit

1. **Linearity Assumption:** House price is linear combination of features
2. **Gaussian Prior:** Weights come from Normal distribution centered at 0
 - Implicit: We believe positive and negative weights equally likely
 - Implicit: We believe all weights independent initially
3. **Gaussian Noise:** Differences between predictions and actual prices are Normally distributed
 - Implicit: Noise has constant variance (homoscedasticity)
 - Implicit: Noise is independent across houses
4. **Fixed Features:** House features (size, bedrooms) are known exactly, not random
5. **MAP Estimation:** We want the single "most plausible" weights, not full distribution

Every Distribution Identity Used

1. **Normal Distribution PDF:**

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Used for: Prior distribution of weights, noise distribution

2. Independence Multiplicative Rule:

$$p(w_1, w_2, \dots, w_d) = p(w_1) \times p(w_2) \times \dots \times p(w_d)$$

Used for: Combining prior beliefs about different weights

3. Bayes' Theorem:

$$p(\text{weights}|\text{data}) \propto p(\text{data}|\text{weights}) \times p(\text{weights})$$

Used for: Combining prior with likelihood

4. Log Trick:

$$\arg \max p(x) = \arg \max \log p(x)$$

Used because: Products become sums, easier to optimize

Mathematical Tools Checklist

Basic Algebra:

- Summation notation Σ
- Squaring numbers and vectors
- Matrix multiplication (dot products)

Probability Concepts:

- Normal (Gaussian) distribution
- Probability Density Function (PDF)
- Independence assumption
- Maximum Likelihood Estimation (MLE) concept

Optimization:

- Finding maximum/minimum of functions

- Derivatives of quadratic functions
- Concept of "arg min" vs "min"

Statistics:

- Variance (spread measure)
- Bias-variance tradeoff (conceptually)
- Regularization concept

Linear Algebra:

- Vectors as lists of numbers
- Norm: $\|w\|^2 = w_1^2 + w_2^2 + \dots + w_d^2$
- Matrix notation (for advanced version)

The Beautiful Connection

Ridge Regression = **Mathematical implementation** of this philosophical stance:

"Extraordinary claims require extraordinary evidence."

The λ controls how much evidence we require before believing in large weight values. The MAP interpretation tells us **why** this mathematical form makes sense from first principles of probability theory.

TL;DR: Ridge Regression is the optimal way to combine data with the sensible default belief that "effects are probably small unless proven otherwise."

Go over the assignment again, and identify derivation heavy concepts. For example, walk through the MAP estimation derivation for W step-by-step. For each line, explain why the step is valid. Do not compress steps. Annotate each operation with the identity, theorem, or fact used.

Detailed Step-by-Step Derivations for Assignment

Core Concepts

I. MAP ESTIMATION DERIVATION FOR RIDGE REGRESSION

Problem Setup:

We want to show that the MAP estimate for W given:

- Prior: $W \sim \mathcal{N}(\mathbf{0}, I_d)$
- Likelihood: $Y|W \sim \mathcal{N}(XW, \lambda I_n)$
- Observed data: $Y = \mathbf{y}$

Gives us the ridge regression objective.

Step-by-Step Derivation:

Line 1: Start with Bayes' Theorem

$$p(W|\mathbf{y}) = \frac{p(\mathbf{y}|W)p(W)}{p(\mathbf{y})}$$

Why valid: Fundamental theorem of probability relating conditional, joint, and marginal distributions.

Line 2: Focus on proportionality (since $p(\mathbf{y})$ doesn't depend on W):

$$p(W|\mathbf{y}) \propto p(\mathbf{y}|W)p(W)$$

Why valid: For MAP, we only need to find W that maximizes posterior. Constants don't affect argmax.

Line 3: Write the prior PDF using Normal distribution formula:

$$p(W) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}W^T I_d W\right)$$

Why valid: PDF of $\mathcal{N}(\mathbf{0}, I_d)$:

- Mean $\mathbf{0}$
- Covariance $I_d \rightarrow$ precision matrix = I_d
- General form: $p(x) \propto \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$

Line 4: Simplify the prior:

$$p(W) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\|W\|_2^2\right)$$

Why valid: $W^T I_d W = W^T W = \sum_{i=1}^d w_i^2 = \|W\|_2^2$

This is the squared Euclidean norm.

Line 5: Write the likelihood PDF:

$$p(\mathbf{y}|W) = \frac{1}{(2\pi\lambda)^{n/2}} \exp\left(-\frac{1}{2\lambda}(\mathbf{y} - XW)^T(\mathbf{y} - XW)\right)$$

Why valid: PDF of $\mathcal{N}(XW, \lambda I_n)$:

- Mean XW
- Covariance $\lambda I_n \rightarrow \Sigma^{-1} = \frac{1}{\lambda} I_n$
- The exponent: $-\frac{1}{2}(\mathbf{y} - XW)^T(\lambda I_n)^{-1}(\mathbf{y} - XW)$

Line 6: Multiply prior and likelihood:

$$p(\mathbf{y}|W)p(W) = \frac{1}{(2\pi\lambda)^{n/2}(2\pi)^{d/2}} \exp\left(-\frac{1}{2\lambda}\|\mathbf{y} - XW\|_2^2 - \frac{1}{2}\|W\|_2^2\right)$$

Why valid: Product of exponentials: $e^a \cdot e^b = e^{a+b}$

Constants multiply together.

Line 7: Take logarithm (for easier maximization):

$$\log p(W|\mathbf{y}) = \text{constant} - \frac{1}{2\lambda} \|\mathbf{y} - XW\|_2^2 - \frac{1}{2} \|W\|_2^2$$

Why valid: $\log(AB) = \log A + \log B$

$\log(e^x) = x$

Constants become additive constants.

Line 8: Multiply by -2λ (doesn't change argmax):

$$-2\lambda \log p(W|\mathbf{y}) = \text{constant} + \|\mathbf{y} - XW\|_2^2 + \lambda \|W\|_2^2$$

Why valid: Multiplying by positive constant preserves ordering:

If $f(W) > f(W')$ then $cf(W) > cf(W')$ for $c > 0$.

Line 9: Since maximizing posterior = minimizing negative log posterior:

$$W_{\text{MAP}} = \arg \min_W [\|\mathbf{y} - XW\|_2^2 + \lambda \|W\|_2^2]$$

Why valid:

- Maximizing $p(W|\mathbf{y})$ = maximizing $\log p(W|\mathbf{y})$
- Maximizing $\log p(W|\mathbf{y})$ = minimizing $-\log p(W|\mathbf{y})$
- Constants ignored in argmin

Final: This is exactly the ridge regression objective (Equation 1).

II. VECTOR CALCULUS DERIVATION: $\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T A \mathbf{x}) = \mathbf{x}^T (A + A^T)$

Step-by-Step:

Line 1: Write the quadratic form explicitly:

$$f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j$$

Why valid: Definition of matrix multiplication: $(AB)_{ij} = \sum_k A_{ik} B_{kj}$

Line 2: Take partial derivative with respect to x_k :

$$\frac{\partial f}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j$$

Why valid: Derivative is linear operator: $\frac{\partial}{\partial x_k} \sum (\dots) = \sum \frac{\partial}{\partial x_k} (\dots)$

Line 3: Only terms where $i = k$ or $j = k$ contribute:

$$\frac{\partial f}{\partial x_k} = \sum_{j=1}^n A_{kj} x_j + \sum_{i=1}^n x_i A_{ik}$$

Why valid: Product rule + chain rule:

- Term 1: $\frac{\partial}{\partial x_k} (x_k A_{kj} x_j) = A_{kj} x_j$ when $i = k$
- Term 2: $\frac{\partial}{\partial x_k} (x_i A_{ik} x_k) = x_i A_{ik}$ when $j = k$
- Terms with neither $i = k$ nor $j = k$ have derivative 0

Line 4: Recognize these as matrix products:

$$\sum_{j=1}^n A_{kj} x_j = (A\mathbf{x})_k$$

$$\sum_{i=1}^n x_i A_{ik} = (A^T \mathbf{x})_k \quad \text{since } A_{ki}^T = A_{ik}$$

Why valid: Definition of matrix-vector multiplication: $(A\mathbf{x})_i = \sum_j A_{ij} x_j$

And: $\mathbf{x}^T A$ as a row vector has entries $\sum_i x_i A_{ij}$

Line 5: Combine:

$$\frac{\partial f}{\partial x_k} = (A\mathbf{x})_k + (A^T \mathbf{x})_k = ((A + A^T)\mathbf{x})_k$$

Why valid: Vector addition is component-wise: $(u + v)_k = u_k + v_k$

Line 6: Write as row vector:

$$\frac{\partial f}{\partial \mathbf{x}} = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right] = \mathbf{x}^T (A + A^T)$$

Why valid: The gradient (row vector) has entries $\frac{\partial f}{\partial x_k}$

And: $((A + A^T)\mathbf{x})^T = \mathbf{x}^T (A + A^T)^T = \mathbf{x}^T (A^T + A)$

Line 7: Since $A + A^T$ is symmetric:

$$\mathbf{x}^T (A + A^T)^T = \mathbf{x}^T (A^T + A) = \mathbf{x}^T (A + A^T)$$

Why valid: Transpose of sum = sum of transposes: $(A + B)^T = A^T + B^T$

III. RIDGE REGRESSION SOLUTION VIA NORMAL EQUATIONS

Line 1: Start with objective:

$$J(W) = \|\mathbf{y} - XW\|_2^2 + \lambda\|W\|_2^2$$

Why valid: Definition of ridge regression

Line 2: Expand the norms:

$$J(W) = (\mathbf{y} - XW)^T(\mathbf{y} - XW) + \lambda W^T W$$

Why valid: $\|\mathbf{v}\|_2^2 = \mathbf{v}^T \mathbf{v}$ for any vector $\mathbf{v}$

Line 3: Expand the first term:

$$= \mathbf{y}^T \mathbf{y} - \mathbf{y}^T XW - (XW)^T \mathbf{y} + (XW)^T (XW) + \lambda W^T W$$

Why valid: Distributive property: $(a - b)^T(a - b) = a^T a - a^T b - b^T a + b^T b$

Line 4: Simplify (since $\mathbf{y}^T XW$ is scalar, equals its transpose):

$$= \mathbf{y}^T \mathbf{y} - 2\mathbf{y}^T XW + W^T X^T XW + \lambda W^T W$$

Why valid: For scalar a , $a = a^T$, so $\mathbf{y}^T XW = (XW)^T \mathbf{y}$

Line 5: Take derivative with respect to W (using $\frac{\partial}{\partial W}(W^T M W) = 2MW$ for symmetric M):

$$\frac{\partial J}{\partial W} = -2X^T \mathbf{y} + 2X^T XW + 2\lambda W$$

Why valid:

- $\frac{\partial}{\partial W}(\mathbf{y}^T XW) = X^T \mathbf{y}$ (from part 2a)

- $\frac{\partial}{\partial W}(W^T M W) = (M + M^T)W$, and $X^T X$ symmetric
- $\frac{\partial}{\partial W}(\lambda W^T W) = 2\lambda W$

Line 6: Set derivative to zero:

$$-2X^T \mathbf{y} + 2X^T X W + 2\lambda W = 0$$

Why valid: First-order necessary condition for minimum (if convex)

Line 7: Divide by 2:

$$-X^T \mathbf{y} + X^T X W + \lambda W = 0$$

Why valid: Linear scaling of equation

Line 8: Factor W :

$$X^T X W + \lambda I W = X^T \mathbf{y}$$

Why valid: $\lambda W = \lambda I W$ where I is identity matrix

Line 9: Combine terms:

$$(X^T X + \lambda I)W = X^T \mathbf{y}$$

Why valid: Distributive property: $AX + BX = (A + B)X$

Line 10: Solve for W (assuming invertible):

$$W = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$$

Why valid: Multiply both sides by inverse: $A^{-1}Ax = A^{-1}b \Rightarrow x = A^{-1}b$

Note: $X^T X + \lambda I$ always invertible for $\lambda > 0$ (positive definite)

IV. RELU ELBOW LOCATION UPDATE

Part 1: Elbow Definition

Line 1: Single ReLU: $\phi(x) = \max(0, wx + b)$

Why valid: Definition of ReLU activation

Line 2: Elbow is where argument crosses zero:

$$wx + b = 0$$

Why valid: Definition: point where function changes from zero to linear

Line 3: Solve for x :

$$e = -\frac{b}{w} \quad \text{for } w \neq 0$$

Why valid: Basic algebra: $wx = -b \Rightarrow x = -b/w$

Part 2: Gradient Update Effect

Line 4: Gradient descent update:

$$w' = w - \eta \frac{\partial \ell}{\partial w}, \quad b' = b - \eta \frac{\partial \ell}{\partial b}$$

Why valid: Definition of gradient descent with learning rate η

Line 5: Partial derivatives (from chain rule):

$$\frac{\partial \ell}{\partial w} = \underbrace{(\phi(x) - y)}_{\delta} \cdot \underbrace{x}_{\partial z / \partial w} \cdot \underbrace{\mathbf{1}_{wx+b>0}}_{\phi'(z)}$$

$$\frac{\partial \ell}{\partial b} = \delta \cdot 1 \cdot \mathbf{1}_{wx+b>0}$$

Why valid:

- $\ell = \frac{1}{2}(\phi(x) - y)^2$
- $\frac{d\ell}{d\phi} = \phi(x) - y$ (power rule + chain rule)
- $\frac{\partial \phi}{\partial z} = \mathbf{1}_{z>0}$ (derivative of ReLU)
- $\frac{\partial z}{\partial w} = x, \frac{\partial z}{\partial b} = 1$ (linear function derivatives)

Line 6: New elbow location:

$$e' = -\frac{b'}{w'} = -\frac{b - \eta\delta\mathbf{1}_{z>0}}{w - \eta\delta x\mathbf{1}_{z>0}}$$

Why valid: Substitute updates into elbow formula $e = -b/w$

Line 7: Special case analysis (from problem):

If $wx + b \leq 0$ ($\phi(x) = 0$), then $\mathbf{1}_{z>0} = 0$, so:

$$e' = -\frac{b - 0}{w - 0} = -\frac{b}{w} = e$$

Why valid: Indicator function = 0 when ReLU inactive, gradients = 0, no update

KEY IDENTITIES AND THEOREMS USED:

Probability:

1. **Bayes' Theorem:** $P(A|B) = P(B|A)P(A)/P(B)$
2. **Normal PDF:** $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
3. **Multivariate Normal PDF:** $p(\mathbf{x}) \propto \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right)$
4. **Independence:** $p(x_1, \dots, x_n) = \prod_i p(x_i)$ for independent variables

Linear Algebra:

1. **Transpose properties:** $(AB)^T = B^T A^T, (A + B)^T = A^T + B^T$
2. **Norm equivalence:** $\|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x}$
3. **Matrix inverse:** $A^{-1}A = I$ when invertible
4. **Symmetric matrices:** $A = A^T$

Calculus:

1. **Chain rule:** $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$
2. **Product rule:** $\frac{d}{dx}(uv) = u'v + uv'$
3. **Derivative of quadratic:** $\frac{d}{dx} x^2 = 2x$
4. **Derivative of linear:** $\frac{d}{dx}(wx + b) = w$
5. **Partial derivatives:** Treat other variables as constants